

Optimization of the Bit Mapping for LDPC-Coded Faster-Than-Nyquist Systems

Jiayi Yang^{1b}, Shuangyang Li^{1b}, *Member, IEEE*, Qianfan Wang^{1b}, *Member, IEEE*,
Xiao Ma^{1b}, *Member, IEEE*, and Giuseppe Caire^{2b}, *Fellow, IEEE*

Abstract—This letter focuses on the analysis and optimization of the bit mapping for low-density parity-check (LDPC) coded faster-than-Nyquist (FTN) systems with high-order modulation. We propose the extrinsic information transfer (EXIT) chart analysis for LDPC-coded FTN systems based on the Ungerboeck observation model, where the vector-input and vector-output mutual information of the FTN detector is approximated using a suitably trained neural network (NN). Leveraging the EXIT chart, we optimize the bit mapping, i.e., the assignment of coded bits to the binary labeled constellation points, specifically for FTN systems. Through threshold analysis and simulations, it is shown that for an FTN system using a variant of the LDPC codes specified in the 5G standard, coded bits with highly reliable positions (information bits) in the Tanner graph preferentially to be transmitted over bit-wise sub-channels with higher capacity, while low-degree bits preferentially to be transmitted over bit-wise sub-channels with lower capacity. Numerical results show that: 1) the LDPC-coded FTN system with the optimized mapping outperforms those with random mappings, consistent with the proposed EXIT chart analysis; 2) under the same spectral efficiency, the LDPC-coded FTN system with optimized mapping also outperforms its 5G LDPC-coded Nyquist counterpart.

Index Terms—Extrinsic information transfer (EXIT) chart, faster-than-Nyquist (FTN), neural network (NN).

I. INTRODUCTION

FASTER-THAN-NYQUIST (FTN) signaling was formally introduced and analyzed by Mazo in 1975 with sinc pulses [1], demonstrating that symbols can be transmitted

at rates up to about 25% above the Nyquist limit without reducing the minimum Euclidean distance. Applying FTN acceleration to other practical pulse shapes or combining it with modern coding techniques appeared roughly two to three decades later, marking the beginning of the modern development of coded FTN systems [2]. Owing to these advantages, FTN signaling has since attracted significant interest in various communication scenarios, including satellite [3], optical [4], and wireless communications [5].

However, FTN signaling inherently introduces inter-symbol interference (ISI). To alleviate the impact of ISI in iterative-recovery systems, two effective solutions are tailored to FTN signaling, the design of dedicated channel codes and the optimization of bit mapping. For channel codes, a variety of low-density parity-check (LDPC) codes have been proposed, including quasi-cyclic (QC) LDPC codes with increased girth [6] and uniformly shortened LDPC codes that better accommodate the ISI [7]. More recently, a class of improved 5G LDPC codes optimized for FTN signaling has been proposed in [8] based on extrinsic information transfer (EXIT) chart analysis, where the mutual information (MI) function of the FTN detector is characterized using polynomial fitting techniques.

For high-order modulation, bit-mapping optimization can provide notable gains in conventional Nyquist systems. However, extending such optimization to FTN signaling is challenging, as the detector's input-output space grows exponentially with modulation order, making polynomial fitting impractical. To address this, we employ a neural network (NN) to approximate the vector MI of the FTN detector, enabling EXIT chart analysis under different bit-mapping strategies. The main contributions of this letter are listed as follows: Distinguished from [8], we use an NN to approximate the vector-input and vector-output MI function of the FTN detector to characterize the MI of the bit-wise sub-channels induced by bit-interleaved coded modulation (BICM) with high-order modulations. This allows us to develop the EXIT chart analysis tailored for FTN signaling with high-order modulation. By the EXIT chart analysis, we optimize the bit mapping of LDPC coded FTN system. Decoding thresholds and results reveal that, for the LDPC coded FTN system, the coded bits with a lower degree in the Tanner graph are preferentially placed to be transmitted over sub-channels with lower capacity. Although demonstrated for FTN signaling, the NN-assisted EXIT framework is detector-agnostic and can be applied to other BICM systems with soft-in/soft-out detection.

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Jiayi Yang and Xiao Ma are with the School of Computer Science and Engineering and Guangdong Key Laboratory of Information Security Technology, Sun Yat-sen University, Guangzhou 510006, China (e-mail: yangji256@mail2.sysu.edu.cn; maxiao@mail.sysu.edu.cn).

Shuangyang Li and Giuseppe Caire are with the Electrical Engineering and Computer Science, Technical University of Berlin, 10587 Berlin, Germany (e-mail: shuangyang.li@tu-berlin.de; caire@tu-berlin.de).

Qianfan Wang is with the Department of Computer Science, City University of Hong Kong, Hong Kong, SAR, China (e-mail: qwang742@cityu.edu.hk).

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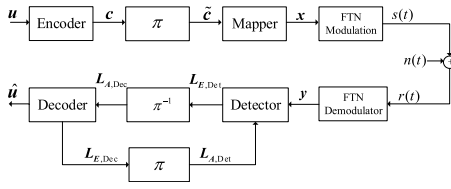


Fig. 1. The diagram of the LDPC-coded FTN system.

II. PRELIMINARIES

A. System Model

The diagram of the LDPC-coded FTN system is shown in Fig. 1. At the transmitter, the information sequence $\mathbf{u} \in \mathbb{F}_2^k$ is first encoded by the encoder of the LDPC code $\mathcal{C}[n, k]$, resulting in a codeword $\mathbf{c} \in \mathbb{F}_2^n$. Then, the codeword $\mathbf{c}$ is interleaved by an interleaver π into $\tilde{\mathbf{c}}$, which is rewritten as $\tilde{\mathbf{c}} = (\tilde{c}_0, \tilde{c}_1, \dots, \tilde{c}_{N-1})$ with $\tilde{c}_t \in \mathbb{F}_2^m$ and $N = n/m$ (for simplicity, assuming that n/m is integer). By the mapper, $\tilde{\mathbf{c}}$ is mapped to the symbol vector $\mathbf{x}$ with $x_t = \phi(\tilde{c}_t)$ and $\mathbf{x}$ is then modulated by the FTN modulator, resulting in

$$s(t) = \sum_{\ell=0}^{N-1} x_\ell h(t - \ell\tau T), \quad (1)$$

where x_ℓ denotes the ℓ -th energy-normalized symbol, $h(t)$ is the shaping pulse, T is the Nyquist symbol period, and $\tau \in (0, 1]$ is the FTN compression factor [1].

Assume that the FTN signal $s(t)$ is transmitted over an AWGN channel, yielding the received signal $r(t) = s(t) + n(t)$, where $n(t)$ is a noisy signal with one-sided power spectral density (PSD) N_0 . Then, the discrete received sequence $\mathbf{y}$ is obtained by first matched-filtering of $r(t)$ and then sampling at the FTN rate [9], which can be represented by $\mathbf{y} = \mathbf{G}\mathbf{x} + \boldsymbol{\eta}$, based on the Ungerboeck observation model, where $\mathbf{G}$ is a Toeplitz matrix characterizing the ISI among transmitted symbols [10]. The matrix $\mathbf{G}$ is constructed from the ISI coefficients g_i , where $-L \leq i \leq L$, and each coefficient satisfies

$$g_i = \int_{-\infty}^{\infty} h(t)h^*(t - i\tau T)dt, \quad -L \leq i \leq L. \quad (2)$$

The term L denotes the ISI truncation length, where only the $2L + 1$ ISI taps centered at g_0 are assumed to have significant energy, while the remaining taps with negligible energy are set to zero for simplicity. The diagram of the receiver of the LDPC-coded FTN system is shown in Fig. 1, where Turbo-equalization receiver structure is adopted. The estimated information sequence $\hat{\mathbf{u}}$ is obtained by performing outer iterations up to $J_{\max}$ between the detector and decoder, and inner iterations up to $I_{\max}$ for LDPC codes. For Ungerboeck observation model, we apply the sum-product algorithm (SPA) for detection, as it has demonstrated promising performance with low computational complexity [11].

B. Block-Wise Interleaver for High-Order Modulation

For conventional BICM systems [12], a random interleaver is typically employed. However, due to the varying reliabilities of different variable nodes in irregular codes, optimizing the interleaver design may provide potential performance gains for FTN signaling, as confirmed by results in conventional

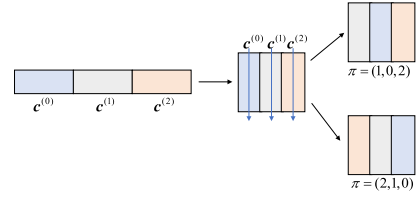


Fig. 2. Illustration of the bit-to-symbol mapping via block-wise interleaving, where each column permutation corresponds to a different mapping pattern.

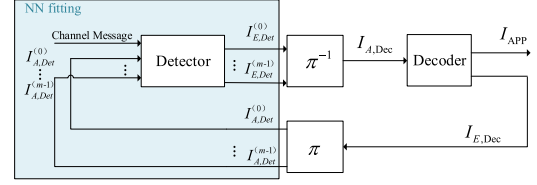


Fig. 3. The diagram of the MI update for the LDPC-coded FTN system, where the extrinsic MI is exchanged between the detector and the decoder.

BICM systems with higher-order modulations [13], [14]. To simplify the description of bit mapping, we rewrite and divide the codeword $\mathbf{c}$ of length n into m segments as $\mathbf{c} = (\mathbf{c}^{(0)}, \mathbf{c}^{(1)}, \dots, \mathbf{c}^{(m-1)})$, where each segment $\mathbf{c}^{(i)} = (c_0^{(i)}, c_1^{(i)}, \dots, c_{n/m-1}^{(i)})$ for $0 \leq i \leq m-1$. We then apply a block-wise interleaver as shown in Fig. 2, the codeword is arranged into an array of dimension $n/m \times m$ and the interleaving is applied on the columns, i.e., the interleaved array corresponds to a permutation of the columns then each row of the array forms a label of a signal point in a constellation of size $M = 2^m$. For example, in Fig. 2, different mapping patterns such as $\pi = (0, 1, 2)$ and $\pi = (1, 0, 2)$ correspond to $x_t = \phi(c_t^{(0)}, c_t^{(1)}, c_t^{(2)})$ and $x_t = \phi(c_t^{(1)}, c_t^{(0)}, c_t^{(2)})$, respectively.

III. LDPC-CODED MODULATION FOR FASTER-THAN-NYQUIST SIGNALING

In this section, we present the EXIT analysis for the LDPC-coded FTN system with high-order modulation. The diagram of the EXIT analysis is illustrated in Fig. 3. Let $\mathbf{I}_{A, \text{Det}} = (I_{A, \text{Det}}^{(0)}, \dots, I_{A, \text{Det}}^{(m-1)})$ and $\mathbf{I}_{E, \text{Det}} = (I_{E, \text{Det}}^{(0)}, \dots, I_{E, \text{Det}}^{(m-1)})$ denote the *a priori* and extrinsic MI of the FTN detector, respectively. The input MI to the i -th parallel branch of the detector is expressed as $\mathbf{I}_{A, \text{Det}}^{(i)} = (I_{A, \text{Det}}^{(i,0)}, \dots, I_{A, \text{Det}}^{(i, n_b/m-1)})$, and the output MI is $\mathbf{I}_{E, \text{Det}}^{(i)} = (I_{E, \text{Det}}^{(i,0)}, \dots, I_{E, \text{Det}}^{(i, n_b/m-1)})$, for $0 \leq i \leq m-1$, assuming n_b/m is an integer. Similarly, we represent the input and output MI vectors of the LDPC decoder as $\mathbf{I}_{A, \text{Dec}} = (I_{A, \text{Dec}}^0, \dots, I_{A, \text{Dec}}^{n_b-1})$ and $\mathbf{I}_{E, \text{Dec}} = (I_{E, \text{Dec}}^0, \dots, I_{E, \text{Dec}}^{n_b-1})$, respectively. The extrinsic MI from the decoder can be rearranged as $\mathbf{I}_{E, \text{Dec}} = (I_{E, \text{Dec}}^{(0)}, \dots, I_{E, \text{Dec}}^{(m-1)})$, where each segment is defined by $\mathbf{I}_{E, \text{Dec}}^{(i)} = (I_{E, \text{Dec}}^{(i,0)}, \dots, I_{E, \text{Dec}}^{(i, n_b/m-1)})$ with $I_{E, \text{Dec}}^{(i, \ell)} = I_{E, \text{Dec}}^{in_b/m + \ell}$, for $0 \leq i \leq m-1$ and $0 \leq \ell \leq n_b/m-1$.

A. EXIT Chart Analysis for the LDPC Codes

In this letter, we focus on protograph-based LDPC codes, as 5G LDPC codes are constructed by protograph. Given the

input *a priori* MI $I_{A,\text{Dec}}$, we can update the output MI $I_{E,\text{Dec}}$ for the protograph-based LDPC code. Precisely, considering the base matrix $\mathbf{B}$ of a protograph-based LDPC code as

$$\mathbf{B} = \begin{bmatrix} b_{0,0} & b_{0,1} & \cdots & b_{0,n_b-1} \\ b_{1,0} & b_{1,1} & \cdots & b_{1,n_b-1} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m_b-1,0} & \cdots & \cdots & b_{m_b-1,n_b-1} \end{bmatrix}, \quad (3)$$

we define the notations for the protograph-based EXIT (PEXIT) analysis of LDPC codes: $I_{Ev}^{i,j}$ denotes the extrinsic MI from the j -th variable node (VN) to the i -th check node (CN); $I_{Ec}^{i,j}$ denotes the extrinsic MI from the i -th CN to the j -th VN; and I_{App}^j denotes the *a posteriori* MI of the j -th VN, which is used for convergence check. The MI update is summarized below, where $J(\cdot)$ and $J^{-1}(\cdot)$ are given in [15].

- *Initialization:* For $0 \leq i \leq m_b - 1, 0 \leq j \leq n_b - 1$, set $I_{Ev}^{i,j} = I_{Ec}^{i,j} = 0$.

- *Extrinsic MI update from the j -th VN to the i -th CN:*

$$I_{Ev}^{i,j} = J \left(\sqrt{\sum_{l \neq j} b_{l,j} \left[J^{-1}(I_{Ec}^{l,j}) \right]^2 + [J^{-1}(I_{A,\text{Dec}}^j)]^2} \right). \quad (4)$$

- *Extrinsic MI update from the i -th CN to the j -th VN:*

$$I_{Ec}^{i,j} = 1 - J \left(\sqrt{\sum_{l \neq j} b_{i,l} \left[J^{-1}(1 - I_{Ev}^{i,l}) \right]^2} \right). \quad (5)$$

- *Calculate *a posteriori* probability (APP) MI and extrinsic MI:* $I_{\text{App}}^j = J \left(\sqrt{[J^{-1}(I_{Ev}^j)]^2 + [J^{-1}(I_{A,\text{Dec}}^j)]^2} \right)$,

$$\text{where } I_{E,\text{Dec}}^j = J \left(\sqrt{\sum_l b_{l,j} [J^{-1}(I_{Ec}^{l,j})]^2} \right).$$

- *Convergence check:* If $|I_{\text{App}}^j - 1| < \delta$ for all $j < n_b - m_b$, decoding is declared successful; otherwise, it fails.

B. MI Approximation Via Neural Networks for FTN Detector

For 2^m -quadrature amplitude modulation (QAM) with a given labeling, there are m modulation bit sub-channels. At the ℓ -th invocation of the detector, the input and output MI vectors are $(I_{A,\text{Det}}^{(0,\ell)}, \dots, I_{A,\text{Det}}^{(m-1,\ell)})$ and $(I_{E,\text{Det}}^{(0,\ell)}, \dots, I_{E,\text{Det}}^{(m-1,\ell)})$, respectively. To model the mapping from input MI and SNR to output MI, we incorporate E_b/N_0 and use a neural network (NN) to approximate the MI transfer function of the FTN detector.

We adopt a fully connected multilayer perceptron (MLP) that takes a $(m+1)$ -dimensional input and outputs an m -dimensional MI vector. A training dataset of size D is generated via Monte Carlo simulation, where $\mathbf{X} \in \mathbb{R}^{(m+1) \times D}$ stacks the input MI and E_b/N_0 samples, and $\hat{\mathbf{Y}} \in \mathbb{R}^{m \times D}$ stacks the corresponding detector output MI samples. The NN is defined as

$$\mathbf{Y} = f(\mathbf{X}) = f_{\text{norm}}(\mathbf{h}_{L-1}), \quad (6)$$

with the hidden-layer recursion

$$\mathbf{h}_0 = \mathbf{X}, \quad \mathbf{h}_l = f_{\text{act}}(\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l), \quad l = 1, \dots, L-1, \quad (7)$$

where $f_{\text{act}}(\cdot)$ and $f_{\text{norm}}(\cdot)$ denote the activation and output-normalization functions, respectively. The network is trained using the mean squared error (MSE) loss between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$.

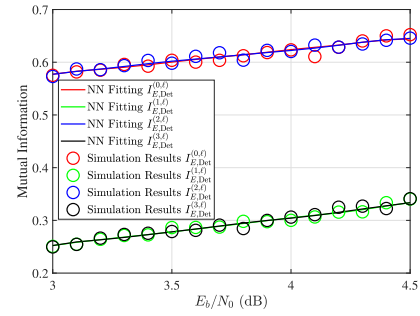


Fig. 4. The NN fitting results and the actual results with $L_m = 4$ and $\tau = \frac{2}{3}$, where the prior MI for the detector is set as $I_{A,\text{Det}} = \mathbf{0}$.

As an example, we consider an FTN system with a T -orthogonal root-raised cosine (RRC) pulse ($\beta = 0.3$), 16-QAM (Gray labeling), and $\tau = 2/3$. A four-layer MLP with two hidden layers of sizes 30 and 20 is used to approximate $I_{E,\text{Det}}$ given $I_{A,\text{Det}} = \mathbf{0}$ and E_b/N_0 . Fig. 4 shows that the NN-based fitting closely matches the Monte Carlo results for all bit levels. Moreover, for Gray-labeled QAM, the m bit sub-channels naturally cluster into $m/2$ reliability classes, which in turn affects the convergence behavior of the decoder.

Remark: This letter adopts Gray labeling and optimizes only the mapping between coded bits and modulation bit positions. Other labelings, such as anti-Gray and set-partitioning, were also tested under both FTN and Nyquist signaling but showed consistently inferior performance, likely due to their mismatch with the 5G LDPC code structure.

C. EXIT Analysis for LDPC-Coded High-Order FTN System

Based on the vector-input and vector-output MI updates for the detector and decoder, the EXIT chart analysis for LDPC-coded FTN system with high-order modulation can be established, see Fig. 3 for reference. Given the channel message and $I_{E,\text{Dec}}$ of length n_b produced by the decoder, the detector is invoked n_b/m times. At the ℓ -th invocation, the input MI of length m to the detector is constructed as

$$(I_{A,\text{Det}}^{(0,\ell)}, \dots, I_{A,\text{Det}}^{(m-1,\ell)}) = \pi \left(I_{E,\text{Dec}}^{(0,\ell)}, \dots, I_{E,\text{Dec}}^{(m-1,\ell)} \right), \quad (8)$$

where π denotes the block-wise interleaver defined in Sec. II. Similarly, at the ℓ -th invocation, the output MI of the detector is deinterleaved to produce the input for the decoder, which is denoted as

$$(I_{A,\text{Dec}}^{(0,\ell)}, \dots, I_{A,\text{Dec}}^{(m-1,\ell)}) = \pi^{-1} \left(I_{E,\text{Det}}^{(0,\ell)}, \dots, I_{E,\text{Det}}^{(m-1,\ell)} \right). \quad (9)$$

After completing all n_b/m detector invocations, the *a priori* MI $I_{A,\text{Dec}}$ for the decoder is obtained. For clarity, the overall EXIT chart procedure is summarized in Algorithm 1. Finally, a bisection search is performed to find the minimum E_b/N_0 required for successful decoding, which is referred to as the decoding threshold of the LDPC-coded FTN system. Note that the proposed scheme appears only in the *offline* design, where the NN-assisted EXIT analysis selects a fixed mapping for each base graph and modulation. This mapping is implemented as a simple interleaver/deinterleaver, so the online complexity and impact on the 5G NR specification are negligible.

Algorithm 1 EXIT Analysis for Coded High-Order FTN

Input: E_b/N_0 , π , B , $J_{\max}$, $I_{\max}$, δ .
Initialization: Set $I_{E,\text{Det}} = I_{A,\text{Det}} = I_{A,\text{Dec}} = I_{E,\text{Dec}} = 0$.
for $i = 0$ **to** $J_{\max} - 1$ **do**
 for $\ell = 0$ **to** $n_b/m - 1$ **do**
 Take $(I_{E,\text{Dec}}^{(0,\ell)}, \dots, I_{E,\text{Dec}}^{(m-1,\ell)})$ as input to obtain $(I_{A,\text{Det}}^{(0,\ell)}, \dots, I_{A,\text{Det}}^{(m-1,\ell)})$ as (8).
 Take $(I_{A,\text{Det}}^{(0,\ell)}, \dots, I_{A,\text{Det}}^{(m-1,\ell)})$ and E_b/N_0 as input and calculate the extrinsic MI of detector based on NN.
 Take $(I_{E,\text{Det}}^{(0,\ell)}, \dots, I_{E,\text{Det}}^{(m-1,\ell)})$ as input to obtain $(I_{A,\text{Dec}}^{(0,\ell)}, \dots, I_{A,\text{Dec}}^{(m-1,\ell)})$ as (9).
 for $t = 0$ **to** $I_{\max} - 1$ **do**
 • Update the MI between VNs and CNs using $I_{A,\text{Dec}}^j$ ($0 \leq j \leq n_b - 1$) according to (4) and (5).
 • Calculate $I_{A,\text{PP}}^j$ and $I_{E,\text{Dec}}^j$ for $0 \leq j \leq n_b - 1$.
 if $|I_{A,\text{PP}}^j - 1| < \delta$, $\forall j < n_b - m_b$ **then**
 Report decoding success and return.

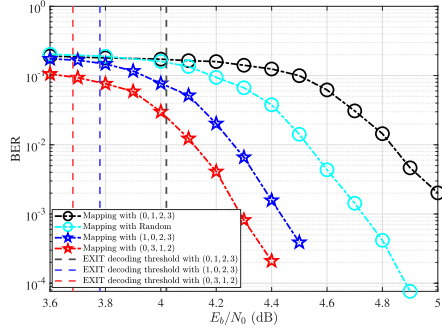


Fig. 5. The thresholds and performance for different bit mapping π , where the improved 5G LDPC code with $n = 15488$ and $r = 1/2$ is employed.

D. Optimized Mapping Results for the LDPC-Coded FTN

Example 1 (Performance and Decoding Thresholds for Different π): Consider the LDPC-coded FTN system with $\tau = 2/3$ and 16-QAM, where the improved 5G LDPC code with $n = 15488$ and $r = 1/2$ is employed [16]. Four different bit mapping (interleaving) patterns are considered: 1) random interleaving (no structure between coded bits and modulation bits); 2) mapping pattern with $\pi = (0, 3, 1, 2)$; 3) mapping pattern with $\pi = (1, 0, 2, 3)$; 4) mapping pattern with $\pi = (0, 1, 2, 3)$. The decoding thresholds (via the proposed EXIT chart analysis) and performance (via simulations) are shown in Fig. 5. It can be observed that properly chosen mappings lead to improved performance, and the performance ranking is consistent with the thresholds obtained from the EXIT analysis, thereby validating the effectiveness of the proposed EXIT analysis. In particular, the optimized mapping pattern of $\pi = (0, 3, 1, 2)$ achieves about a 0.4 dB gain over the random interleaver at a BER of 10^{-3} . Intuitively, for the improved 5G LDPC-coded FTN system, coded bits with highly reliable positions (information bits) in the Tanner graph preferentially to be transmitted over sub-channels with higher capacity, while low-degree bits preferentially to be transmitted over sub-channels with lower capacity as $\pi = (0, 3, 1, 2)$. Note that for 16-QAM with Gray labeling, although there are $4! = 24$ permutations, symmetry reduces them to six

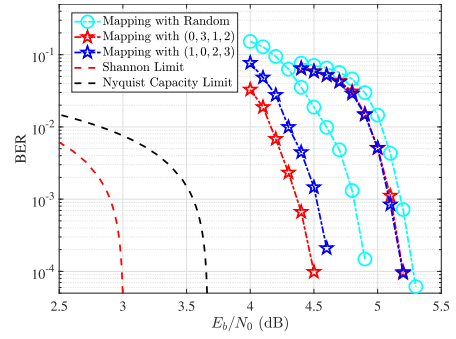


Fig. 6. The performance of the FTN and Nyquist system at 3 bits/s with a 30% roll-off RRC pulse (about 2.3 bits/s/Hz), where the improved 5G LDPC codes with $r = 1/2$ and $n = 11264$ are employed for the FTN, while original LDPC codes with $r = 3/4$ and $n = 11264$ are utilized for Nyquist.

MI-distinct orderings. We evaluated all six and report a representative subset for clarity.

Example 2 (Performance Comparison with Nyquist Signaling): Consider the LDPC-coded FTN system with $\tau = 2/3$ and 16-QAM modulation. For comparison, we also evaluate an LDPC-coded Nyquist system with the same modulation. In this example, three bit mappings are considered for both systems: i) Nyquist-optimized mapping $(1, 0, 2, 3)/(0, 3, 1, 2)$ obtained via one-dimensional, ii) FTN-optimized mapping $(0, 3, 1, 2)$ by the proposed EXIT chart, iii) randomly interleaved mapping. To ensure the same SE of 3 bits per channel use, the FTN system employs a rate-1/2 improved 5G LDPC code [16] with $n = 11264$, while the Nyquist system uses a rate-3/4 standard 5G LDPC code with the same $n = 11264$. The performance comparison is shown in Fig. 6. It is observed that the FTN system with the optimized mapping $(0, 3, 1, 2)$ outperforms that with Nyquist-optimized mapping. Compared to random interleaving, mapping optimization of the FTN system provides larger gains than that in Nyquist systems. Notably, under the same SE, the optimized FTN system achieves a gain of about 0.7 dB over all Nyquist counterparts at $\text{BER} = 10^{-3}$. At a rate of 3 bits/s with a 30% roll-off RRC pulse (about 2.3 bits/s/Hz), our proposed FTN signaling combined with LDPC coding operates about 1.5 dB away from the Shannon limit at $\text{BER} = 10^{-4}$ [17], while the Nyquist-based 5G LDPC scheme remains around 2.2 dB away. These observations also confirm that our approach achieves about 0.5 dB improvement over the simple-coding-plus-FTN schemes reported in [18].

Remark: For fair comparison, we fix the total transmitted energy per codeword. Both FTN and Nyquist signals use sequences of comparable length with equal average symbol energy, ensuring identical total energy. Under this setup, FTN signaling can yield an improved spectral efficiency as demonstrated in [2], because of its improved constrained capacity compared to the Nyquist counterpart. Similar fairness criteria—same block length and equal spectral efficiency—have also been adopted in [6] and [19]. Moreover, with the same shaping pulse $h(t)$, their power spectral densities are proportional to $|H(f)|^2$, occupying the same bandwidth—clarifying that FTN does not inherently expand bandwidth.

Example 3 (Performance Comparison for Non-Product Constellation): Consider an LDPC-coded FTN system with

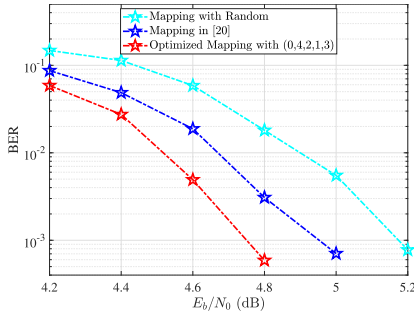


Fig. 7. The BER with different mapping schemes for 32-QAM, using the improved 5G LDPC code of length $n = 7040$ and rate $r = 1/2$.

TABLE I

NN-OPTIMIZED MAPPING SCALABILITY ACROSS SYSTEM SETTINGS

Mapping Pattern	$\beta = 0.4$	$n = 3872$	$r = 1/3$	Fading	$\tau = 0.77$
Random	4.22	5.30	3.1	6.1	> 3.8
(0, 1, 2, 3)	> 4.4	> 5.4	2.75	> 6.2	3.7
(1, 0, 2, 3)	3.98	4.92	> 3.8	5.98	3.52
(0, 3, 1, 2)	3.90	4.70	2.50	5.90	3.62

$\tau = 0.8$ and 32-QAM modulation, using the improved 5G LDPC code with $n = 7040$ and $r = 1/2$. Three mapping patterns are evaluated: i) random interleaving (no structure between coded bits and modulation bits); ii) the optimized mapping $\pi = (0, 4, 2, 1, 3)$ obtained by our proposed EXIT chart analysis; and iii) the heuristic mapping from [20], where systematic bits are assigned to relatively more reliable modulation positions. We note that this heuristic approach does not account for the specific code structure nor provide decoding thresholds, and thus cannot guarantee optimal performance. As shown in Fig. 7, the optimized mapping achieves about a 0.2 dB gain over the heuristic scheme of [20], confirming the effectiveness of the proposed EXIT-based bit-mapping design.

Example 4 (Robustness of the NN-based EXIT Chart): Consider the NN trained with $\beta = 0.3$, $\tau = 2/3$, and 16-QAM over the AWGN channel, using LDPC codes constructed from 5G NR base graph 1 (BG1) as specified in [21]. The EXIT chart analysis and simulations (with $n \approx 15488$ and $r = 1/2$) indicate that the mapping $\pi = (0, 3, 1, 2)$ consistently achieves the best performance among the considered patterns. To verify scalability, we further simulate the required E_b/N_0 at $\text{BER} = 10^{-3}$ under different mapping patterns for various roll-off factors β , code lengths, code rates, channel conditions, and compression factors τ , as summarized in Table I, where the best-performing results are highlighted in bold. The results show that for different β , code lengths, code rates, and channel conditions, the optimal mapping remains consistent with that obtained under the original training parameters, whereas changes in τ lead to different optimal mappings. These findings demonstrate the scalability of the scheme across β , code lengths, code rates, and channel conditions, while its scalability to different τ values is more limited.

IV. CONCLUSION

We proposed an EXIT chart analysis for LDPC-coded FTN signaling with high-order modulation, where the detector's

MI is approximated using NN. Based on this framework, an optimized bit-mapping scheme was developed, yielding clear performance gains over random interleaving. Under the same spectral efficiency, the improved 5G LDPC-coded FTN system outperforms the 5G LDPC-coded Nyquist counterpart.

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